POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5013 | Semester V | Theory | 4 Credits

Wastewater Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5013 as Wastewater Engineering in Semester V for Civil Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public wastewater-engineering curriculum sources and is not represented as recovered official SITTTR Course 5013 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. Wastewater design/operation requires current local regulations, discharge/reuse criteria, site data, laboratory methods and competent professional review.

Verified source note: The official detailed source was inaccessible. Public sources supporting wastewater characterisation, sewer systems, treatment units, microbiology, sludge and septic tanks are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Wastewater Engineering studies collection, characterisation, treatment and responsible disposal or reuse of wastewater. It connects public health, environmental protection, hydraulics, microbiology and treatment-unit operation.

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Prerequisite foundation

Water-supply/environmental fundamentals, basic chemistry, hydraulics and responsible laboratory/site safety.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain wastewater sources, characteristics and basic analysis terms.
2. Explain sewerage and collection-system concepts.
3. Explain physical, chemical and biological treatment processes.
4. Explain sludge management, onsite sanitation, discharge/reuse and operational safety concepts.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain wastewater characteristics and common quality parameters.	Understand
CO2	Explain collection-system and sewerage planning concepts.	Understand
CO3	Compare primary, secondary and selected advanced treatment processes.	Apply
CO4	Explain sludge, onsite treatment, disposal/reuse and safety considerations.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Wastewater Sources and Characteristics	Sources, flow variation, physical/chemical/biological constituents, BOD, COD, solids, nutrients, microorganisms and sampling awareness.	Approx. 14 hours	CO1	Module 1
2	Collection and Sewerage Systems	Sewer types, appurtenances, hydraulic concepts, infiltration/inflow, pumping and planning.	Approx. 15 hours	CO2	Module 2
3	Treatment Processes and Treatment Trains	Preliminary/primary treatment, sedimentation, biological processes, activated sludge, trickling filters, ponds, aerated lagoons, disinfection and process selection.	Approx. 17 hours	CO3	Module 3
4	Sludge, Onsite Systems and Reuse	Sludge processing/disposal, septic tanks, effluent management, reuse, monitoring and plant safety.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Wastewater Sources and Characteristics

Sources, flow variation, physical/chemical/biological constituents, BOD, COD, solids, nutrients, microorganisms and sampling awareness.

CO1 · Approx. 14 hours

Collection and Sewerage Systems

Sewer types, appurtenances, hydraulic concepts, infiltration/inflow, pumping and planning.

CO2 · Approx. 15 hours

Treatment Processes and Treatment Trains

Preliminary/primary treatment, sedimentation, biological processes, activated sludge, trickling filters, ponds, aerated lagoons, disinfection and process selection.

CO3 · Approx. 17 hours

Sludge, Onsite Systems and Reuse

Sludge processing/disposal, septic tanks, effluent management, reuse, monitoring and plant safety.

CO4 · Approx. 14 hours

MODULE 1

Wastewater Sources and Characteristics

Sources, flow variation, physical/chemical/biological constituents, BOD, COD, solids, nutrients, microorganisms and sampling awareness.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

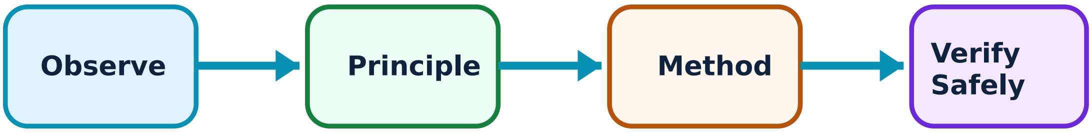
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Wastewater Sources and Characteristics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Wastewater sources and flows−

Domestic, commercial, industrial and storm-related contributions create variable wastewater flows and pollutant loads. Understanding source and variation supports sound collection and treatment planning.

Study and application method

Identify source, likely contaminants, flow pattern and receiving-environment concern; distinguish flow rate from pollutant mass load.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify source, likely contaminants, flow pattern and receiving-environment concern; distinguish flow rate from pollutant mass load.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never assume industrial discharge composition is domestic wastewater; verify through approved monitoring and regulation.

2. Quality parameters−

BOD indicates oxygen demand from biodegradable matter, COD measures chemically oxidisable matter, and suspended solids represent particulate content. Nutrients, pathogens, pH and temperature also matter.

Study and application method

State the parameter, unit, analytical basis and what process/environmental behaviour it represents.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State the parameter, unit, analytical basis and what process/environmental behaviour it represents.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not compare results from different sampling/analysis methods without noting the method and timing.

3. Microbiology and public health−

Microorganisms enable biological treatment but pathogens can create health risk. Process control balances conditions for desired treatment organisms and safe exposure control.

Study and application method

Relate a treatment step to targeted contaminants and state required hygiene, sample handling and PPE practices.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Relate a treatment step to targeted contaminants and state required hygiene, sample handling and PPE practices.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Treat wastewater and sludge as potentially hazardous; follow approved biosafety procedures.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Collection and Sewerage Systems

Sewer types, appurtenances, hydraulic concepts, infiltration/inflow, pumping and planning.

Approx. 15 hours | CO2 | Understand · Apply

Learning goals

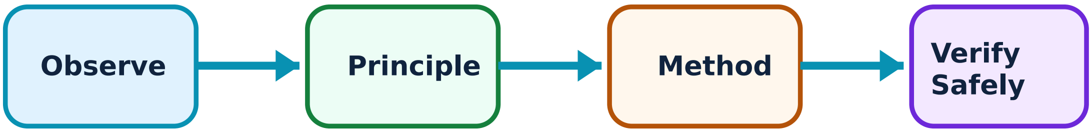
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Collection and Sewerage Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Sewer systems and appurtenances−

Separate, combined and partially separate systems collect/carry wastewater and stormwater under different conditions. Manholes, traps, vents, pumping stations and inspection points support operation.

Study and application method

Select a conceptual system based on flow type, terrain, rainfall, existing infrastructure and environmental risk, while stating data needed for design.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Select a conceptual system based on flow type, terrain, rainfall, existing infrastructure and environmental risk, while stating data needed for design.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Confined-space sewer entry is high risk and requires permits, atmospheric testing, rescue arrangements and trained personnel.

2. Hydraulic performance−

Gravity sewers need suitable slope, flow velocity and capacity to transport solids while avoiding excessive deposition, surcharge or erosion. Pumping may be needed where gravity flow is impractical.

Study and application method

Sketch hydraulic grade/flow direction, list required survey/flow/roughness data and explain self-cleansing versus excessive-velocity concern.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch hydraulic grade/flow direction, list required survey/flow/roughness data and explain self-cleansing versus excessive-velocity concern.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not size sewers from generic rules without approved local standards and population/flow data.

3. Infiltration, inflow and maintenance−

Groundwater infiltration and stormwater inflow can overload collection/treatment systems. Inspection, cleaning, repair and source control maintain capacity and reduce environmental incidents.

Study and application method

Use a basic diagnostic plan: flow pattern, rainfall correlation, CCTV/inspection results and priority repair areas.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a basic diagnostic plan: flow pattern, rainfall correlation, CCTV/inspection results and priority repair areas.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never use unapproved bypasses or discharge untreated sewage to address overload.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Treatment Processes and Treatment Trains

Preliminary/primary treatment, sedimentation, biological processes, activated sludge, trickling filters, ponds, aerated lagoons, disinfection and process selection.

Approx. 17 hours | CO3 | Understand · Apply

Learning goals

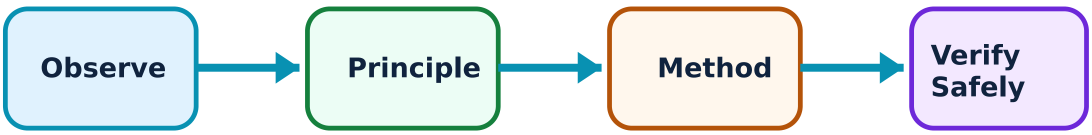
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Treatment Processes and Treatment Trains: identify the system, state its principle, follow the correct method and verify the result safely.

1. Preliminary and primary treatment−

Screens, grit removal and sedimentation protect downstream equipment and remove settleable material. Primary treatment reduces solids and some organic load before biological processing.

Study and application method

Trace influent through units and state objective, main removal mechanism and expected operational issue for each.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace influent through units and state objective, main removal mechanism and expected operational issue for each.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Screenings and grit require safe handling and approved disposal; they are not ordinary waste.

2. Biological treatment−

Activated sludge, trickling filters, ponds and lagoons use microorganisms to transform biodegradable pollutants. Oxygen transfer, biomass, loading, retention time and settling affect performance.

Study and application method

Compare attached-growth versus suspended-growth system, energy/land need, loading sensitivity and typical operational controls.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare attached-growth versus suspended-growth system, energy/land need, loading sensitivity and typical operational controls.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not operate aeration or recycle controls without trained supervision; abrupt changes can destabilise biological treatment.

3. Treatment-train selection−

A treatment train combines unit processes to meet influent characteristics, effluent requirements, site, energy, land, reliability and operator-capacity constraints.

Study and application method

Start with influent/effluent data, identify required removal mechanisms, propose unit sequence and list design/permit checks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Start with influent/effluent data, identify required removal mechanisms, propose unit sequence and list design/permit checks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: No treatment train is universal; local discharge/reuse requirements govern selection.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Sludge, Onsite Systems and Reuse

Sludge processing/disposal, septic tanks, effluent management, reuse, monitoring and plant safety.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

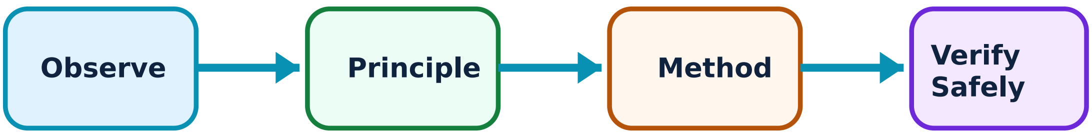
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sludge, Onsite Systems and Reuse: identify the system, state its principle, follow the correct method and verify the result safely.

1. Sludge treatment and management−

Sludge contains concentrated solids and potential pathogens. Thickening, stabilisation/digestion, dewatering and approved final use/disposal reduce volume, odour and risk.

Study and application method

Identify sludge source, likely characteristics, treatment objective, handling controls and final destination/permit requirement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify sludge source, likely characteristics, treatment objective, handling controls and final destination/permit requirement.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never dispose sludge without authorised route, testing and records.

2. Septic tanks and onsite treatment−

Septic tanks provide primary settling and anaerobic treatment for suitable onsite contexts; downstream soil absorption/other treatment and maintenance are vital to prevent failure.

Study and application method

Explain inflow, settling/scum/sludge zones, outlet protection, periodic desludging and site assessment factors.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain inflow, settling/scum/sludge zones, outlet protection, periodic desludging and site assessment factors.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Septic tanks can contain toxic gases and oxygen-deficient atmospheres; entry is prohibited without specialised confined-space controls.

3. Reuse and environmental monitoring−

Treated effluent may be reused when quality, reliability, distribution, exposure control and regulations are satisfied. Monitoring confirms compliance and guides corrective action.

Study and application method

State proposed reuse, target quality parameters, monitoring points/frequency and controls for cross-connection/public exposure.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State proposed reuse, target quality parameters, monitoring points/frequency and controls for cross-connection/public exposure.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Do not reuse effluent without explicit regulatory approval and validated treatment performance.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Wastewater Sources and Characteristics

Use the concept flow to revise observation, principle, method and verification.

Module 2: Collection and Sewerage Systems

Use the concept flow to revise observation, principle, method and verification.

Module 3: Treatment Processes and Treatment Trains

Use the concept flow to revise observation, principle, method and verification.

Module 4: Sludge, Onsite Systems and Reuse

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a wastewater-characteristics table for an instructor-provided scenario.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a conceptual sewerage and treatment flow diagram.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare activated sludge, trickling filter and pond systems for a stated context.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a safety and inspection checklist for a simulated wastewater plant visit.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Wastewater Sources and Characteristics

Explain Wastewater sources and flows and state one correct method, application or precaution.−

Domestic, commercial, industrial and storm-related contributions create variable wastewater flows and pollutant loads. Understanding source and variation supports sound collection and treatment planning.

Answer framework: Identify source, likely contaminants, flow pattern and receiving-environment concern; distinguish flow rate from pollutant mass load.

Important point: Never assume industrial discharge composition is domestic wastewater; verify through approved monitoring and regulation.

Explain Quality parameters and state one correct method, application or precaution.−

BOD indicates oxygen demand from biodegradable matter, COD measures chemically oxidisable matter, and suspended solids represent particulate content. Nutrients, pathogens, pH and temperature also matter.

Answer framework: State the parameter, unit, analytical basis and what process/environmental behaviour it represents.

Important point: Do not compare results from different sampling/analysis methods without noting the method and timing.

Explain Microbiology and public health and state one correct method, application or precaution.−

Microorganisms enable biological treatment but pathogens can create health risk. Process control balances conditions for desired treatment organisms and safe exposure control.

Answer framework: Relate a treatment step to targeted contaminants and state required hygiene, sample handling and PPE practices.

Important point: Treat wastewater and sludge as potentially hazardous; follow approved biosafety procedures.

Module 2 — Collection and Sewerage Systems

Explain Sewer systems and appurtenances and state one correct method, application or precaution.−

Separate, combined and partially separate systems collect/carry wastewater and stormwater under different conditions. Manholes, traps, vents, pumping stations and inspection points support operation.

Answer framework: Select a conceptual system based on flow type, terrain, rainfall, existing infrastructure and environmental risk, while stating data needed for design.

Important point: Confined-space sewer entry is high risk and requires permits, atmospheric testing, rescue arrangements and trained personnel.

Explain Hydraulic performance and state one correct method, application or precaution.−

Gravity sewers need suitable slope, flow velocity and capacity to transport solids while avoiding excessive deposition, surcharge or erosion. Pumping may be needed where gravity flow is impractical.

Answer framework: Sketch hydraulic grade/flow direction, list required survey/flow/roughness data and explain self-cleansing versus excessive-velocity concern.

Important point: Do not size sewers from generic rules without approved local standards and population/flow data.

Explain Infiltration, inflow and maintenance and state one correct method, application or precaution.−

Groundwater infiltration and stormwater inflow can overload collection/treatment systems. Inspection, cleaning, repair and source control maintain capacity and reduce environmental incidents.

Answer framework: Use a basic diagnostic plan: flow pattern, rainfall correlation, CCTV/inspection results and priority repair areas.

Important point: Never use unapproved bypasses or discharge untreated sewage to address overload.

Module 3 — Treatment Processes and Treatment Trains

Explain Preliminary and primary treatment and state one correct method, application or precaution.−

Screens, grit removal and sedimentation protect downstream equipment and remove settleable material. Primary treatment reduces solids and some organic load before biological processing.

Answer framework: Trace influent through units and state objective, main removal mechanism and expected operational issue for each.

Important point: Screenings and grit require safe handling and approved disposal; they are not ordinary waste.

Explain Biological treatment and state one correct method, application or precaution.−

Activated sludge, trickling filters, ponds and lagoons use microorganisms to transform biodegradable pollutants. Oxygen transfer, biomass, loading, retention time and settling affect performance.

Answer framework: Compare attached-growth versus suspended-growth system, energy/land need, loading sensitivity and typical operational controls.

Important point: Do not operate aeration or recycle controls without trained supervision; abrupt changes can destabilise biological treatment.

Explain Treatment-train selection and state one correct method, application or precaution.−

A treatment train combines unit processes to meet influent characteristics, effluent requirements, site, energy, land, reliability and operator-capacity constraints.

Answer framework: Start with influent/effluent data, identify required removal mechanisms, propose unit sequence and list design/permit checks.

Important point: No treatment train is universal; local discharge/reuse requirements govern selection.

Module 4 — Sludge, Onsite Systems and Reuse

Explain Sludge treatment and management and state one correct method, application or precaution.−

Sludge contains concentrated solids and potential pathogens. Thickening, stabilisation/digestion, dewatering and approved final use/disposal reduce volume, odour and risk.

Answer framework: Identify sludge source, likely characteristics, treatment objective, handling controls and final destination/permit requirement.

Important point: Never dispose sludge without authorised route, testing and records.

Explain Septic tanks and onsite treatment and state one correct method, application or precaution.−

Septic tanks provide primary settling and anaerobic treatment for suitable onsite contexts; downstream soil absorption/other treatment and maintenance are vital to prevent failure.

Answer framework: Explain inflow, settling/scum/sludge zones, outlet protection, periodic desludging and site assessment factors.

Important point: Septic tanks can contain toxic gases and oxygen-deficient atmospheres; entry is prohibited without specialised confined-space controls.

Explain Reuse and environmental monitoring and state one correct method, application or precaution.−

Treated effluent may be reused when quality, reliability, distribution, exposure control and regulations are satisfied. Monitoring confirms compliance and guides corrective action.

Answer framework: State proposed reuse, target quality parameters, monitoring points/frequency and controls for cross-connection/public exposure.

Important point: Do not reuse effluent without explicit regulatory approval and validated treatment performance.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Wastewater Sources and Characteristics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Collection and Sewerage Systems.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Treatment Processes and Treatment Trains.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Sludge, Onsite Systems and Reuse.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Sources, flow variation, physical/chemical/biological constituents, BOD, COD, solids, nutrients, microorganisms and sampling awareness..
2. Develop a complete answer or practical plan for: Sewer types, appurtenances, hydraulic concepts, infiltration/inflow, pumping and planning..
3. Develop a complete answer or practical plan for: Preliminary/primary treatment, sedimentation, biological processes, activated sludge, trickling filters, ponds, aerated lagoons, disinfection and process selection..
4. Develop a complete answer or practical plan for: Sludge processing/disposal, septic tanks, effluent management, reuse, monitoring and plant safety..

LAST-MILE REVISION

Quick Revision

Module 1: Wastewater Sources and Characteristics

Sources, flow variation, physical/chemical/biological constituents, BOD, COD, solids, nutrients, microorganisms and sampling awareness.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Collection and Sewerage Systems

Sewer types, appurtenances, hydraulic concepts, infiltration/inflow, pumping and planning.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Treatment Processes and Treatment Trains

Preliminary/primary treatment, sedimentation, biological processes, activated sludge, trickling filters, ponds, aerated lagoons, disinfection and process selection.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Sludge, Onsite Systems and Reuse

Sludge processing/disposal, septic tanks, effluent management, reuse, monitoring and plant safety.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Wastewater sources and flows	Domestic, commercial, industrial and storm-related contributions create variable wastewater flows and pollutant loads.
Quality parameters	BOD indicates oxygen demand from biodegradable matter, COD measures chemically oxidisable matter, and suspended solids represent particulate content.
Microbiology and public health	Microorganisms enable biological treatment but pathogens can create health risk.
Sewer systems and appurtenances	Separate, combined and partially separate systems collect/carry wastewater and stormwater under different conditions.
Hydraulic performance	Gravity sewers need suitable slope, flow velocity and capacity to transport solids while avoiding excessive deposition, surcharge or erosion.
Infiltration, inflow and maintenance	Groundwater infiltration and stormwater inflow can overload collection/treatment systems.
Preliminary and primary treatment	Screens, grit removal and sedimentation protect downstream equipment and remove settleable material.
Biological treatment	Activated sludge, trickling filters, ponds and lagoons use microorganisms to transform biodegradable pollutants.
Treatment-train selection	A treatment train combines unit processes to meet influent characteristics, effluent requirements, site, energy, land, reliability and operator-capacity constraints.
Sludge treatment and management	Sludge contains concentrated solids and potential pathogens.
Septic tanks and onsite treatment	Septic tanks provide primary settling and anaerobic treatment for suitable onsite contexts; downstream soil absorption/other treatment and maintenance are vital to prevent failure.
Reuse and environmental monitoring	Treated effluent may be reused when quality, reliability, distribution, exposure control and regulations are satisfied.

LEARNING RESOURCES

References

Recommended wastewater-engineering references

1. Metcalf & Eddy, Wastewater Engineering: Treatment and Resource Recovery.
2. G. S. Birdie and J. S. Birdie, Water Supply and Sanitary Engineering.
3. Current applicable wastewater discharge/reuse standards.

Approved public wastewater-engineering sources

- https://ocw.mit.edu/courses/1-85-water-and-wastewater-treatment-engineering-spring-2006/
- https://ocw.utm.my/course/view.php?id=275

These sources support the study handbook while official Course 5013 content is unavailable; they do not replace applicable permits, standards or approved plant procedures.